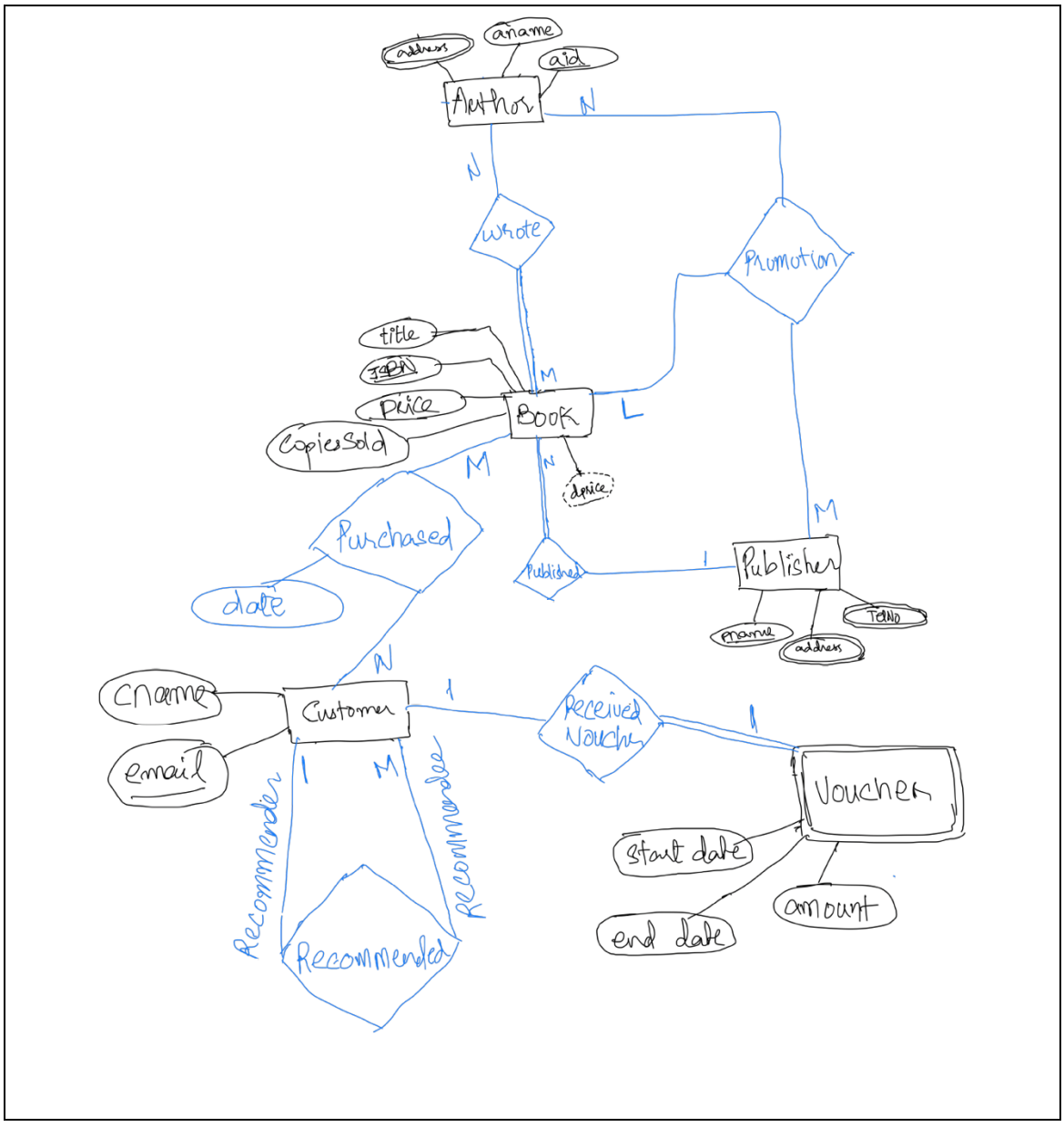
Software Systems

Marks: 60
Time: 2 hrs

Question 1

(a) ER diagram:



(b) Relational Schema:

Book(ISBN, title, price, copiedSold, pname)
Author(aid, aname)
AuthAdress(aid, address)
Publisher(pname)
PubAddress(pname, address)
PubTelephone(pname, TelNo)
Customer(cname, email)

Voucher(customerEmail, startDate, endDate, amount)
Wrote(ISBN, aid)
Purchased(customerEmail, ISBN, date)
Recommended(recommenderEmail, recommndeeEmail)
Promotion(ISBN, aid, pname)

(c)i

```
SELECT name FROM Actor WHERE actorID IN
(
        SELECT DISTINCT actorID FROM Character NATURAL JOIN Play WHERE author = 'Checkov'
        EXCEPT
        SELECT DISTINCT actorID FROM Character NATURAL JOIN Play WHERE author = 'Arthur Miller'
)
```

(c)ii

```
SELECT actorID FROM
(
        SELECT actorID, COUNT(playID) AS pcount FROM
        (
                (SELECT DISTINCT actorID, playID FROM Character)
                NATURAL JOIN
                Play
        )
        WHERE author = 'Shakespeare'
        GROUP BY actorID
)
WHERE pcount >= 3;
```

Question 2

(a)i **TRUE.** Transmission delay is the ratio of packet length to bandwidth: L / R, which is independent of the channel's length.

(a)ii **TRUE.** In circuit-switched networks, where dedicated communication paths are established for the duration of a conversation or data transfer, the concept of queueing delays due to the queuing of data packets is not applicable in the same way as it is in packet-switched networks.

(a)iii **FALSE.** With a variation in load, processing on the nodes will take variable time. However, propagation delay, which is the ratio between distance and propagation speed, will stay the same for the same path.

(b)i **3 RTTs**, 1 RTT for connection establishment, 2 RTTs for base file transfer (window size not sufficient to transfer base file in 1 RTT)
(b)ii **16 RTTs.** 3 RTTs for the base file (as computed in i), 6 RTTs in total for the connections established for the 6 objects, 5 RTTs for the complete transfer of ob1-5, 2 RTTs for the transfer of ob6.
(b)iii **8 RTTs.** 3 RTTs for the base file (as computed in i), 1 RTT for four parallel connections, 1 RTT for ob1-4, 1 RTT for two parallel connections, 1 RTT for ob6 and first 20, 000 bits of ob7, 1 RTT for the remaining 15, 000 bits of ob7.
(note: 7RTTs is also a correct answer. This can happen if the browser chooses to fetch Ob1-3 and Ob6 first, followed by ob4-5).

(b)iv **7 RTTs.** 3 RTTs for the base file (as computed in i). After this, all 6 requests are made at once through the same persistent connection that was used to get the base file. All responses are ready to be sent back together, a total of 80, 000 bits. But with a window size of 20, 000 bits, this will take at least 4 RTTs.

(c) **{k-N+1, k-N+2, …, k}**
We may consider two extreme scenarios. (1) all the ACKS before k have been received at the sender. (2) the sender has not yet received ACK for any of the previous N-1 packets. We will have the following sequence number ranges for the windows in these two scenarios:
(1) [k, k+1, …, K+N-1]
(2) [k-N+1, …, k]
Other window ranges are possible between these two extremes. Therefore, the set of all possible sequence numbers that could be in the window: {k-N+1, k-N+2, …, k}

(d) **151 sec**
Total data to be transferred = 2^{32} bytes
Maximum segment size = 2^9 bytes
Required number of segments= $2^{32} / 2^9 = 2^{23}$
Each segment is attached with a 64 bytes = 2^6 byte header data.

- ⇨ total header data = $2^6 * 2^{23} = 2^{29}$ bytes
- ⇨ Total data to be transmitted = $2^{32} + 2^{29} = 2^3 * 2^{29} + 2^{29} = 9 * 2^{29}$ bytes

Bandwidth = 2^8 Mbps = $2^5 * 10^6$ bytes per sec

Transmission delay = $9 * 2^{29} / 2^5 * 10^6 = 9 * 2^{24} * 10^{-6}$ sec

(an answer up till here will get full credit)

Transmission delay = $9 * 2^{24} * 10^{-6} = 151$ sec

Question 3

(a)i **192.168.0.1** the gateway is the next hop
(a)ii **192.168.0.0 to 192.168.0.127**
The entry in the routing table will not be used for communications with nodes in the computer's own subnet. Based on the mask and the computer's IP, we can detect the following information:
Network ID of this computer: 192.168.0.0 (not usable for hosts)
First usable IP in the subnet: 192.168.0.1
Last usable IP in the subnet: 192.168.0.126
Broadcast Address: 192.168.0.127 (not usable for hosts)
(a)iii
IP sequence for packet to node1: 192.168.0.4, 192.168.0.35
IP sequence for packet to node2: 192.168.0.4, 192.168.0.1 (gateway), 192.168.0.35
router 1 is simply the gateway. node 1 is in our own subnet. node 2 is in a different subnet. To communicate with node 2, we will need to go through the gateway.
(The ARP protocols will be used to find the MAC addresses of node 1 and router 1)
(a)iv IP-to-MAC mappings:
(**192.168.0.35, EB-01-06-00-23-12**)
(**192.168.0.1, F3-00-21-00-13-44**)

(b) **underlines show values updated on each iteration**

Iteration	R0	R1	R2	R3	R4	Transmitters
0	INF	INF	INF	INF	INF	-
1	18, R5	INF	INF	9, R5	1, R5	R5
2	18, R5	16, R3	6, R5	9, R5	1, R5	R0, R3, R4
3	17, R1	9, R2	6, R5	8, R2	1, R5	R1, R2
4	10, R1	9, R2	6, R5	8, R2	1, R5	R0, R1, R3

(c)i **Refer to lecture notes**.
(c)ii **Refer to lecture notes.**